USCIM: Computing-in-Memory With Unipolar Switching SOT-MRAM

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This brief presents an arithmetic paradigm for computing-in-memory (CIM) using unipolar switching spin-orbit-torque magnetic random access memory (SOT-MRAM) devices. The proposed MRAM array comprised two transistors and one SOT magnetic tunnel junction (MTJ) cell and achieves basic Boolean logic operations [AND, XOR, OR, majority (MAJ)] and full-adder (FA) operations. The hybrid spintronics/CMOS simulation results show that the proposed designs improve latency and energy consumption by about 34%–132% and 212%–489% compared with the existing design, respectively. Monte Carlo simulations further prove the robustness of our design for the effectiveness of write/read operations.

***Index Terms*— Computing-in-memory (CIM), latency and energy, spin-orbit-torque magnetic random access memory (SOT-MRAM), unipolar switching.**

I. Introduction

IN THE decades, the increasingly frequent data exchanges between processors and memories have intensified the bottleneck in the Von Neumann architecture. The "memory wall" and "power wall" challenges have significantly impeded the advancement of computer technology. Computing-in-memory (CIM) technology is one of the best solutions to address these limitations by executing computational tasks within the memory. CIM platforms built with non-volatile device technologies can effectively mitigate the frequent data exchanges to reduce data movement overhead and static power consumption. Compared with other nonvolatile memory devices (such as resistive memory, phase change memory, and ferroelectric memory) [1], [2], [3], MRAM is a promising candidate for CIM applications because of its non-volatility, high durability, high density, and low power consumption [4]. To date, spin-transfer torque magnetic random access memory (STT-MRAM) is the predominant device for implementing MRAM-based CIM [5], [6]. However, its incubation delay limits its write speed and results in relatively high write power consumption [7], [8].

To overcome the shortcomings of STT-MRAM, researchers propose spin-orbit-torque magnetic random access memory (SOT-MRAM), which offers high writing speed and low power consumption [9], [10], [11]. Currently, SOT-MRAM often requires external controls, such as electric fields or STT currents, to assist switching, leading to additional power consumption [12]. Recently, the field-free unipolar switching of SOT-MTJ devices has been proposed to achieve ultrafast speed and low-power switching by controlling the ratio of field-like torque to damping-like torque ($\lambda_{FL}/\lambda_{DL}$) [13], [14], [15]. Wu et al. [16] innovatively implemented CIM by leveraging the unipolar switching characteristics of SOT-MRAM. In their work, basic Boolean operations are achieved by demonstrating a basic operation called tunnel resistance switching (TRS) and decomposing conventional logic. However, this approach resulted in increased operational steps and read/write cycles, which limits the speed and increases power consumption.

Based on the SOT magnetic tunnel junction (SOT-MTJ) characteristics, we proposed a 2-transistor-1-SOT (2T1SOT)-based memory array to complete the basic Boolean logic operations. The main contributions of this article are summarized as follows.

1) The unipolar switching characteristics of the SOT-MTJ are investigated, and the 2T1SOT memory cell circuit along with the MRAM architecture is designed.
2) Based on write and read operations, circuit designs for five basic logic operations [AND, XOR, OR, majority (MAJ) and full-adder (FA)] are proposed in a single read/write cycle to reduce the CIM operation steps and memory units.
3) The feasibility of the proposed scheme is verified on a simulation platform. The results show that compared with existing designs, the delay is reduced by 34%–200%, and power consumption is reduced by 212%–489%.

This article is organized into five sections. Section II introduces the new switching characteristics of the SOT-MTJ. Section III describes the memory cell circuit of the 2T1SOT MRAM, as well as the realization and simulation of basic logic operations. Section IV presents the performance evaluation and reliability analysis. Finally, Section V concludes the results.

II. Unipolar Switching SOT-MTJ

The core component of MRAM is the MTJ [4], which consists of three layers: a free layer (FL), a spacer, and a reference layer (RL). The magnetization direction of the FL can be altered to parallel (high-resistance state "R_{AP}" for MTJ) or antiparallel (low-resistance state "R_P" for MTJ) to the RL. The magnetization of the FL can be changed through

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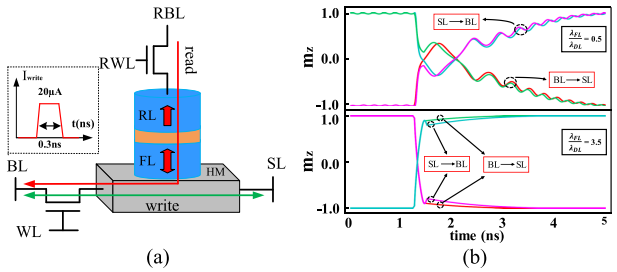


Fig. 1. (a) Schematic of the proposed memory cell. (b) Magnetization switching curves of SOT-MRAM under different amplitudes of pulse currents.

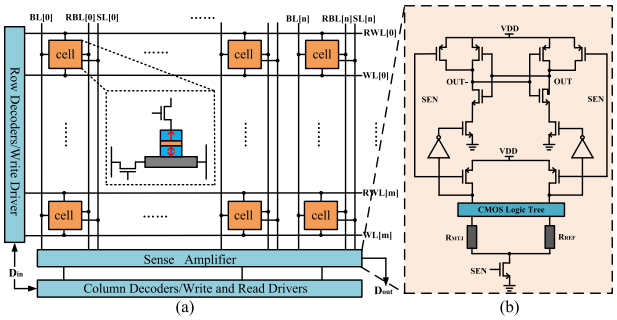


Fig. 2. (a) Structure of 2T1SOT MRAM array. (b) Schematic of the separated precharge sensing amplifier (SPCSA) [22].

various methods, such as magnetic fields, STT, or SOT. SOT-based devices incorporate an additional heavy metal (HM) layer beneath the MTJ in Fig. 1(a). The charge current flowing through this HM layer generates a spin-polarized current, which switches the magnetization of the FL.

The magnetization dynamics of FL can be described by the Landau–Lifshitz–Gilbert–Slonczewski (LLGS) equation

$$\frac{\partial \vec{m}}{\partial t} = -\gamma\mu_0\vec{m}\times\vec{H}_{\text{eff}} + \alpha\vec{m}\times\frac{\partial\vec{m}}{\partial t} - J\xi\lambda_{\text{DL}}\vec{m}\times\left(\vec{m}\times\vec{\sigma}\right) - J\xi\lambda_{\text{FL}}\vec{m}\times\vec{\sigma} \quad (1)$$

where γ is the gyromagnetic ratio, μ_0 is the vacuum magnetic permeability, $\vec{H}_{\text{eff}}$ is the effective field, α is the damping constant, J is the applied current density, and $\vec{m}$ is the unit vector of the FL magnetization. The third and fourth terms on the right side are damping-like torque (λ_{DL}) and field-like torque (λ_{FL}), respectively. In typical SOT switching, the contribution of field-like torque is often neglected. For devices with perpendicular anisotropic SOT-MTJ, an additional field and current are often needed for deterministic switching and the charge current used for SOT is bipolar. Recently, Wang et al. [17] reported the non-zero field-like torque Mo/CoFeB/MgO heterostructures, where the ratio of field-like torque to damping-like torque can be greater than 3. Later on, Wang et al. [18] proposed the possibility of achieving field-free switching by enhancing the ratio of field-like torque and damping-like torque through micromagnetic simulations. Besides, the switched final magnetization state depends only on the amplitude and pulsewidth rather than the polarities of the applied current (namely, achieving unipolar switching). Furthermore, the field-like torque actually affects the chirality of domain walls, and chirality reversal can change the direction of SOT-induced switching. It, therefore, offers more possibilities for realizing the unipolar switching experimentally [19], [20], [21].

Fig. 1(b) shows the results in different ratios of $\lambda_{\text{FL}}/\lambda_{\text{DL}}$. The value m_z on the y-axis represents the direction of FL's magnetization. When $m_z = 1$, the FL's magnetization direction is parallel to the fixed layer's; when $m_z = -1$, it is antiparallel. In the upper panel of Fig. 1(b), when $\lambda_{\text{FL}}/\lambda_{\text{DL}}$ is set to 0.5, the switch of the device is a bipolar process. The magnetizations can only be changed by one of the current directions [either BL to SL or SL to BL in Fig. 1(a)]. When the ratio is set to 3.5, the magnetizations successfully flip regardless of the direction of the current flowing through the HM. The unipolar switching achieves sub-nanosecond (0.3 ns in this simulation) switching speeds through the application of a current pulse with adequate amplitude (~20 μA), offering significant advantages in terms of speed and power consumption.

III. Designs of Logic Operations

Fig. 2(a) shows the architecture of the memory array, which is mainly composed of a decoder, read–write driving circuits, memory cells, and sensing amplifier circuits. Among them, the memory cell is implemented by a 2T1SOT, and the CIM operation of the memory cell can be achieved by activating transistors and the SOT-MTJ through multiple control lines in Fig. 1(a). Basic Boolean logic operations can be performed in a single read/write cycle. Furthermore, leveraging the advantages of unipolar switching characteristics, including low-amplitude pulse driving and sub-nanosecond switching speed, logic operations consume minimal energy. Both of them significantly enhance the overall computational efficiency.

A. AND and XOR Operations

The CIM operation is based on both write and read operations in Fig. 3(a) and (b). For the write operation (green arrow), the word line (WL) is pulled to high, and then the bitline (BL) and source line (SL) are activated to select the memory cell. A current pulse applied to the BL flows through the HM layer of the selected memory cell to the SL, thereby reversing the state of the SOT-MTJ to accomplish the write operation. For the read operation (red arrow), the WL and read WL (RWL) are activated, and then the BL and read BL (RBL) are selected to perform the read operation through the sensing amplifier. In Fig. 2(b), we used a separable pre-charge sense amplifier circuit to read, which is advantageous for high-speed and low-power applications since only dynamic charge or discharge current is involved during the reading process [22].

Using the 2T1SOT-MTJ unit's write and read operations, we can realize the logic operation involving three operands (such as A, B, and C). The operand "A" represents the voltage across the RWL, which controls the transistor above the MTJ. The operand "B" represents the data stored in the MTJ, with high resistance ($m_z = -1$) indicating "0" and low resistance ($m_z = 1$) indicating "1." The operand "C" indicates whether a current pulse is applied to BL. For a logical operation, data B are first stored in the cell with a write operation is performed. If C = 1, a pulse current flows through the HM layer, reversing the MTJ state (B change); if C = 0, the MTJ state remains unchanged. During the read operation, operand A controls the transistor above the MTJ whether turns on or off.

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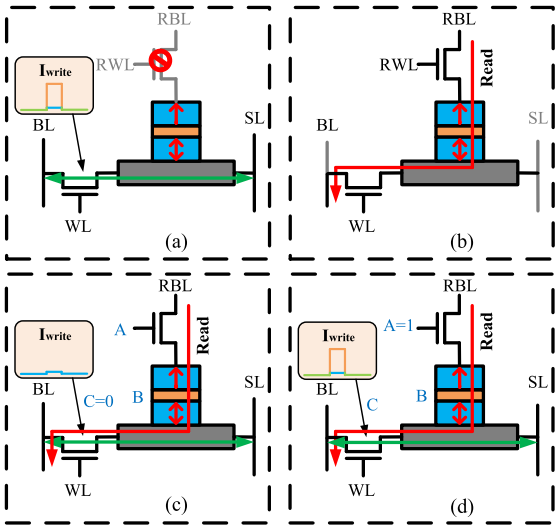


Fig. 3. Schematic of operations. (a) Write operation. (b) Read operation. (c) AND operation. (d) XOR operation.

TABLE I
TRUE TABLE OF THE PROPOSED CIM

A	0	0	0	0	1	1	1	1
B	0	0	1	1	0	0	1	1
C	0	1	0	1	0	1	0	1
OUT	0	0	0	0	0	1	1	0

When A = 1 (the transistor is on), the sensing amplifier compares the MTJ resistance with a reference resistance and outputs the result: "0" for high MTJ resistance and "1" for low MTJ resistance. Conversely, when A = 0 (the transistor is off), the sense amplifier output is "0" (OUT = 0), regardless of the MTJ resistance. Table I presents the truth table for the three operands based on this configuration, leading to the output function

$$\text{OUT} = A(\bar{B}C + B\bar{C}) = A(B \oplus C). \tag{2}$$

Using (2), basic logic operations can be implemented by configuring the inputs. As shown in Fig. 3(c), when C = 0 (no current pulse), OUT = AB realizes the "AND" operation. The "XOR" operation is achieved when A = 1, giving OUT = B ⊕ C, with the transistor always on during the read operation in Fig. 3(d).

B. OR and MAJ Operations

The proposed "OR" operation needs to use two memory cells in the same column. The operands "A" and "B" are written into MTJ0 and MTJ1 initially, for example. Subsequently, the transistors above the MTJs will be turned on together to perform the read operation. By setting an appropriate reference resistance, the result of the "OR" operation can be read out correctly.

For the implementation of the "MAJ" operation, the traditional logical expression needs to be reconfigured

$$\text{MAJ} = AB + AC + BC = A(B \oplus C) + BC. \tag{3}$$

It has been shown that a combination of three-input MAJ operations can represent any Boolean function containing three variables. Therefore, efficient implementation of MAJ logic in memory can improve the computational efficiency of the system. The "MAJ" operation can be realized using

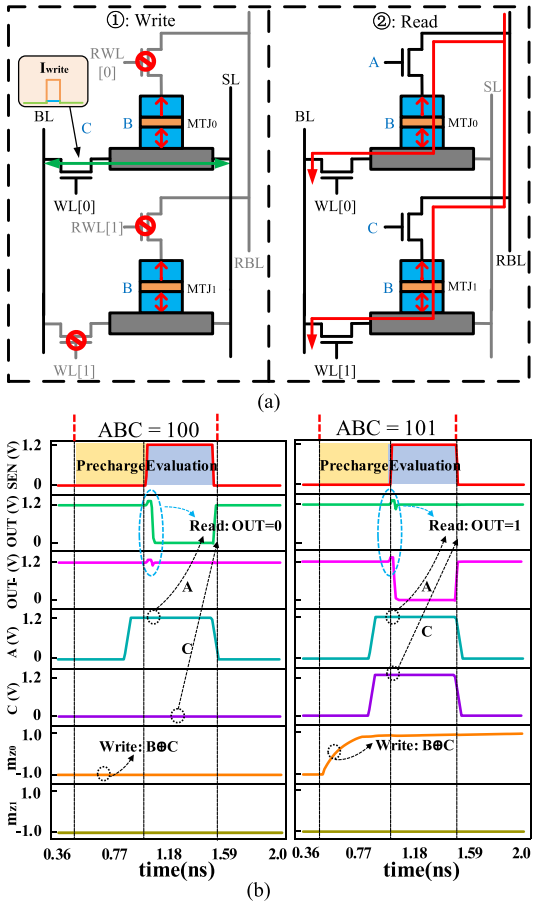


Fig. 4. (a) Schematic of OR operation. (b) Transient simulation of MAJ operation with values of "100" and "101."

two memory cells according to (3). As shown in Fig. 4(a), the operand "B" is stored in both MTJ0 and MTJ1. A write operation is then performed on MTJ0 with the control signal of "C." During the subsequent read operation, the RWL [0] voltage for MTJ0 depends on operand "A," while the voltage of RWL [1] for MTJ1 will be decided by the value of "C." MTJ0 implements the operation of "A (B ⊕ C)" and MTJ1 achieves the function of "BC." The result of the "MAJ" operation is obtained when the states of the two memory cells are read out simultaneously.

Fig. 4(b) provides the transient simulation results when "ABC = 100" and "ABC = 101," respectively. We take the latter case as an example. Initially, "B = 1" is stored in both MTJ0 and MTJ1 ($m_z = 1$). Then a write operation is performed on MTJ0 to switch its state, while the state of MTJ1 remains unchanged. After that the read operation is performed turning on RWL [0] ("A = 1") and RWL [1] ("C = 1"), with a result of the output "OUT = 1."

The selection of reference resistance (R_{REF}) is critical during logic operations. For "AND" and "XOR" operations, R_{REF} should be set between R_P and R_{AP}. For the "OR" and "MAJ" operations, the total resistance of the two branches must be considered. As mentioned previously, when the reading branch of a memory cell is turned off, the output of the sensing amplifier is always "0." During this state, the current flowing through the memory cell is negligible, resulting in a high-resistance state (R_H, which is much larger than R_{AP}). Therefore, the data

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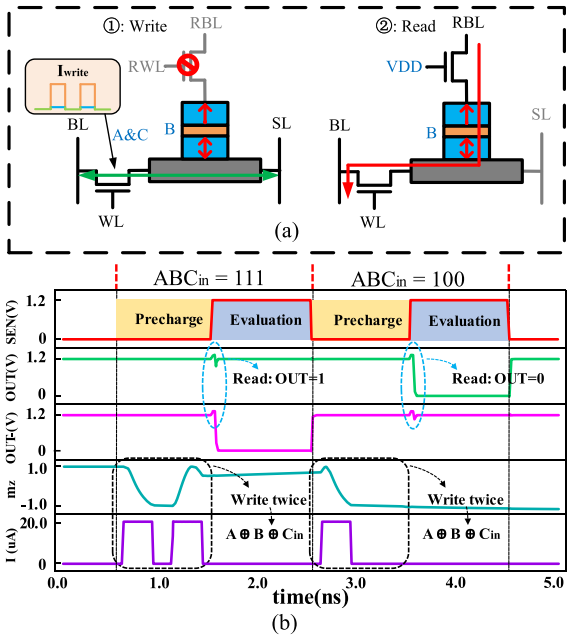


Fig. 5. Schematic of "SUM" operation. (b) Transient simulation of "SUM" operation with values of "111" and "100."

on this path are assumed to be "0," regardless of whether the MTJ state is parallel or antiparallel. For the "OR" operation, the total resistance has three cases: $R_P \| R_P$ $(1+1)$, $R_P \| R_{\mathrm{AP}}$ $(0+1)$, and $R_{\mathrm{AP}} \| R_{\mathrm{AP}}$ $(0+0)$. Thus, the setting is $R_P \| R_P < R_P \| R_{\mathrm{AP}} < R_{\mathrm{REF}} < R_{\mathrm{AP}} \| R_{\mathrm{AP}}$. For the "MAJ" operation, there are five cases for the total resistance: $R_P \| R_P$ $(1+1)$, $R_P \| R_{\mathrm{AP}}$ $(0+1)$, $R_P \| R_H$ $(0+1, \approx R_P)$, $R_{\mathrm{AP}} \| R_{\mathrm{AP}}$ $(0+0)$, and $R_{\mathrm{AP}} \| R_H$ $(0+0, \approx R_{\mathrm{AP}})$. Accordingly, we should set $R_P \| R_P < R_P \| R_{\mathrm{AP}} < R_P < R_{\mathrm{REF}} < R_{\mathrm{AP}} \| R_{\mathrm{AP}} < R_{\mathrm{AP}}$. The R_{REF} range for the "MAJ" operation is narrower than that of other operations. When performing a "MAJ" operation, a small detection margin may lead to reading errors due to device process variations. Detailed reliability analysis is discussed in Section IV.

C. 1-bit FA Operation

The addition operation is a fundamental arithmetic function in a processing system [23]. The logical functions of a 1-bit FA can be expressed as follows:

$$\mathrm{SUM} = A \oplus B \oplus C_{\mathrm{in}} \tag{4}$$

$$C_{\mathrm{out}} = \mathrm{MAJ}(A, B, C_{\mathrm{in}}) \tag{5}$$

where "A" and "B" are the addends, "C_{in}" is the input carry, "SUM" is the output sum, and "C_{out}" is the output carry. We demonstrate the implementation of a 1-bit FA using three memory cells. One memory cell performs "XOR" operations twice to obtain "SUM," while the other two memory cells perform "MAJ" operations to output "C_{out}." To obtain the result of "A ⊕ B ⊕ C_{in}," the operand "B" is first written to a memory cell shown in Fig. 5(a). Subsequently, write operations are performed twice controlled by operand "A" and operand "C_{in}." The first write operation represents "A ⊕ B" and the second results in "A ⊕ B ⊕ C_{in}." Finally, a read operation is performed to obtain "SUM." By calculating the "SUM" and carrying separately in different columns, a 1-bit full addition operation can be completed in just one read–write cycle, significantly reducing the operation time.

TABLE II
DEVICE PARAMETERS

Parameters	Value
Heavy Mental Dimension	30 nm × 60 nm × 5 nm
MTJ Dimension [24]	20 nm × 20 nm
Resistance area product [24]	1e11 Ω·m²
Saturation magnetization (M_s) [25]	1e6 A/m
Resistivity of heavy mental width [25]	2e-6 Ω/m
Oxide Layer Thickness (t_{ox})	0.85 nm
Free Layer Thickness (t_{sl}) [25]	0.7 nm
Damping Constant (α) [25]	0.03
Ratio between SOT field-like torque and damping-like torque ($\lambda_{FL}/\lambda_{DL}$) [16]	3.5
Tunel Magnettoresistance (TMR) [24]	2.4

In Fig. 5(b), the transient simulation results for "SUM" when "$\mathrm{ABC_{in}} = 111$" and "$\mathrm{ABC_{in}} = 100$." For the operation of "$\mathrm{ABC_{in}} = 111$," "B = 1" is stored in the MTJs initially, followed by two write operations on the MTJs. The magnetization of the MTJ is reversed twice due to the application of two current pulses controlled by "A = 1" and "$\mathrm{C_{in}} = 1$." Finally, a read operation is performed to retrieve the result of "1."

IV. PERFORMANCE EVALUATION

The hybrid MTJ/CMOS circuit is simulated using the 40-nm CMOS design kit and the SOT-MRAM compact model [18], [31], [32], [33]. The device parameters of the SOT-MRAM are provided in Table II. We set the supply voltage to 1.2 V, the write current pulse amplitude to 20 μA, and the current duration to 0.3 ns.

Various operations including write/read, AND, XOR, OR, MAJ, and FA are simulated to acquire circuit-level delay and power consumption metrics. For write operation, due to unipolar switching characteristics of SOT-MRAM, the latency and energy for writing "1" and "0" are the same at 0.36 ns and 0.86 fJ/bit, respectively. Since the read latency of the MRAM is asymmetric, we consider the result for the worst case energy consumption at 6.13fJ/bit. Detailed results for computing operations are presented in Table III. Thanks to the ultra-fast switching speed and low write current of the SOT-MRAM, data can be written with a high speed and low power consumption. These metrics are then compared with those reported in other literature [3], [7], [12], [16], [27], [28], [30], as summarized in Table III. Notably, compared with other designs, our design exhibits a minimum 34% and 212% reduction in delay and power consumption, respectively. Furthermore, with minimal changes to peripheral circuits, our design achieves basic logic operations using only one or two memory cells, and 1-bit full addition using only three logic cells, thus improving area efficiency. For "AND," "OR," and "MAJ" operations, a write-back operation is required to store the result, introducing additional energy overhead. This aspect warrants further attention and optimization in future iterations.

Furthermore, using the Destiny memory simulation tool, a 1-MB RAM with a 64B data length is evaluated. Table IV presents the performance evaluation results of the proposed array and compares them with previous works. Compared with previous works, the proposed design achieves a 72% reduction in energy consumption of AND, OR operations while the latency is comparable as well. For MAJ and FA operations,

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TABLE III

COMPARISON OF THE PROPOSED DESIGN WITH OTHER CIM DESIGNS

Operation	Device	latency (ns)	energy (fJ)
AND	STT-MRAM [7]	6	323.5
	TD-STT-MRAM [26]	**0.36**	29.98
	US-SOT-MRAM [16]	0.6	39.26
	SOT-MRAM [27]	2	230.2
	SHE-IMTJ [28]	0.78	81
	This work	0.38	**6.99**
OR	STT-MRAM [7]	6	109.5
	US-SOT-MRAM [16]	0.6	26.17
	STT-IMC [29]	1.837	181.6
	SOT-MRAM [27]	2	209.17
	SHE-IMTJ [28]	0.73	79.5
	This work	**0.39**	**8.38**
XOR	STT-MRAM [7]	10	278.9
	TD-STT-MRAM [26]	0.68	61.69
	SOT/GAA-CNTFET [30]	1.8	59
	SOT-MRAM [27]	4	384.9
	EE-VGSOT [8]	0.3	14.78
	This work	**0.37**	**6.22**
MAJ	US-SOT-MRAM [16]	1.0	61.99
	STT-IMC [29]	0.762	73.86
	EE-VGSOT [8]	3.3	227.6
	This work	**0.43**	**10.51**
FA	US-SOT-MRAM [16]	1.3	68.13
	STT-IMC [29]	1.847	373
	TD-STT-MRAM [26]	**0.85**	103
	SOT/GAA-CNTFET [30]	8	370
	VSH-MRAM [12]	4.33	77.35
	This work	0.97	**18.41**

TABLE IV

COMPARISON RESULTS IN 1-MB ARRAY LEVEL

Operation	Device	latency (ns)	energy (pJ)
AND	STT-MRAM [7]	8.717	238.337
	US-SOT-MRAM [16]	1.743	86.58
	This work	**1.60**	**49.9**
OR	STT-MRAM [7]	8.717	128.769
	US-SOT-MRAM [16]	1.543	79.21
	This work	1.56	**46.82**
MAJ	US-SOT-MRAM [16]	2.143	94.69
	FeRAM [3]	3.281	95.6
	This work	**1.74**	**76.39**
FA	US-SOT-MRAM [16]	2.443	105.674
	This work	**2.20**	123.13

the performance of latency and energy is improved by at least 23%. However, the write-back phase associated with AND, OR operations resulted in additional overhead, which is a potential direction for future optimization.

To validate the robustness of the proposed scheme, we performed a reliability analysis. Regarding write reliability, process variations in the SOT-MTJ can potentially induce write errors, particularly when the write pulse time and the interval between two writes are too short, as in the case of the "SUM" operation. To assess this, we conducted 1000 Monte Carlo simulations, considering a 3% process variation in parameters of t_{ox}, t_{sl}, and TMR of the MTJ. The result demonstrates that the magnetization direction consistently flips correctly and steadily under two write pulses, affirming the reliability of the write operation in Fig. 6(a). For the read operation, process variations can diminish the detection margin of the detection amplifier. Among the operations, the "MAJ" operation is particularly susceptible to interference due to the narrow margin for setting the reference resistor. To evaluate this, Monte Carlo simulations were performed for these two cases to obtain the output results in Fig. 6(b). When the total resistance is less (higher) than R_{REF}, "OUT" is successfully

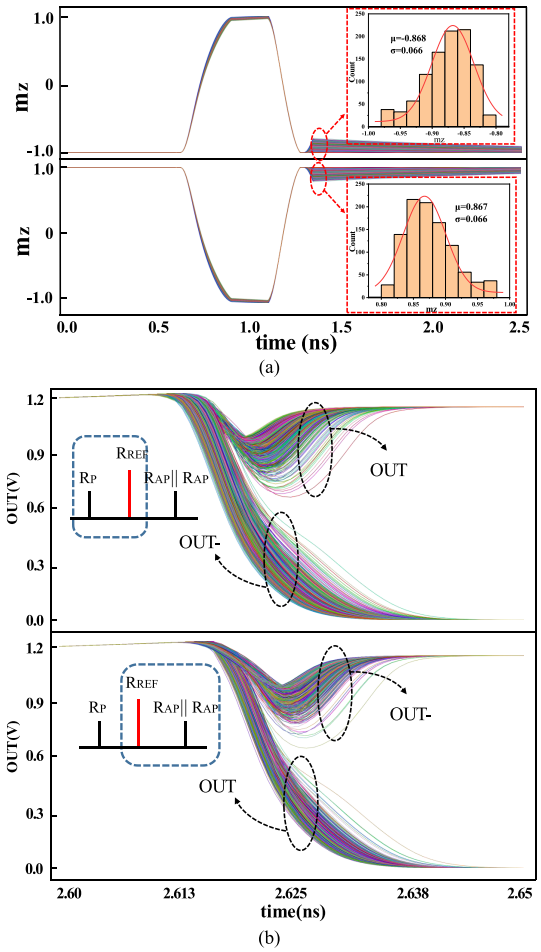


Fig. 6. (a) Transient simulation of two write operations with initial values of "0" and "1." (b) Monte Carlo simulation result of read operations with the total resistance of R_P and $R_{AP} \| R_{AP}$.

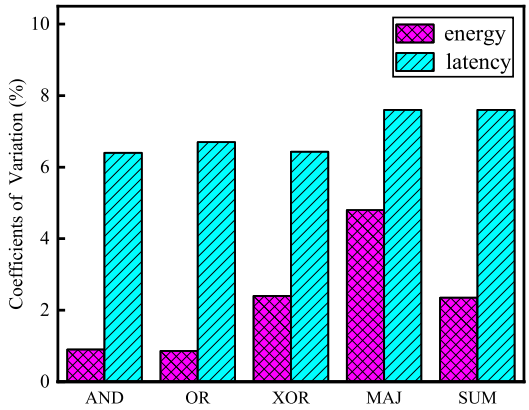


Fig. 7. Effect of the process variation on the performance of the proposed circuit.

pulled up to VDD (GND) while "OUT-" is pulled down to GND (VDD). For the single-bit read operation, all the cases are performed correctly considering the process variations of the device. The sensing amplifier's robust performance ensures the generation of correct results [22].

The same 1000 Monte Carlo simulations were performed to further characterize the impact of process variations on the circuit performance of basic logic operations. The coefficient of variation σ/μ, where μ represents the mean and σ represents the standard deviation, was used to characterize

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the performance of the proposed scheme [9], [34]. As shown in Fig. 7, the circuit performance of all the logic operations exhibits variations below 10%, indicating the robustness of the proposed design to process variations.

V. Conclusion

In this article, we introduce a novel 2T1SOT-based CIM architecture and implement various logic operations, including AND, XOR, OR, MAJ, and 1-bit FA leveraging the unipolar switching characteristics of SOT-MRAM. By configuring inputs for write and read operations, basic five logic operations can be accomplished in a single read/write cycle, thereby enhancing operational speed and efficiency. Simulation results validate the efficiency of our proposed scheme, showcasing a remarkable 34%–132% reduction in delay and 212%–489% reduction in power consumption compared with existing designs. Furthermore, Monte Carlo simulations show the robust reliability of our proposed design during read-and-write operations and the performance of the process variation. Our findings underscore the potential of MRAM-based CIM platforms and provide valuable insights for future research and development endeavors in this domain.

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References

[1] J. Reuben and S. Pechmann, "Accelerated addition in resistive RAM array using parallel-friendly majority gates," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 29, no. 6, pp. 1108–1121, Jun. 2021.

[2] M. Bertuletti, I. M. Noz-Martín, S. Bianchi, A. G. Bonfanti, and D. Ielmini, "A multilayer neural accelerator with binary activations based on phase-change memory," *IEEE Trans. Electron Devices*, vol. 70, no. 3, pp. 986–992, Mar. 2023.

[3] Q. Huang et al., "Computing-in-memory using ferroelectrics: From single- to multi-input logic," *IEEE Des. Test. IEEE Des. Test. Comput.*, vol. 39, no. 2, pp. 56–64, Feb. 2022.

[4] P. Barla, V. K. Joshi, and S. Bhat, "Spintronic devices: A promising alternative to CMOS devices," *J. Comput. Electron.*, vol. 20, no. 2, pp. 805–837, Apr. 2021.

[5] B. Jahannia, S. A. Ghasemi, and H. Farbeh, "An energy efficient multi-retention STT-MRAM memory architecture for IoT applications," *IEEE Trans. Circuits Syst. II: Exp. Briefs*, vol. 71, no. 3, pp. 1431–1435, Mar. 2024.

[6] B. Jahannia, S. A. Ghasemi, and H. Farbeh, "Multi-retention STT-MRAM architectures for IoT: Evaluating the impact of retention levels and memory mapping schemes," *IEEE Access*, vol. 12, pp. 26562–26580, 2024.

[7] H. Zhang, W. Kang, K. Cao, B. Wu, Y. Zhang, and W. Zhao, "Spintronic processing unit in spin transfer torque magnetic random access memory," *IEEE Trans. Electron Devices*, vol. 66, no. 4, pp. 2017–2022, Apr. 2019.

[8] M. R. Sarkar and C. Y. Yi, "An in-memory computing architecture utilizing energy-efficient VGSOT MRAM device," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 7, pp. 3258–3262, Jul. 2024.

[9] A. Mohseni, M. H. Moaiyeri, A. Amirany, and M. Hadi Rezayati, "Protecting the intellectual property of binary deep neural networks with efficient spintronic-based hardware obfuscation," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 7, pp. 3146–3156, Jul. 2024.

[10] M. BahmanAbadi, A. Amirany, and M. H. Moaiyeri, "Nonvolatile quaternary pre-charged MRAM cell design based on emerging technologies," in *Proc. 32nd Int. Conf. Electr. Eng. (ICEE)*, May 2024, pp. 1–5.

[11] Z. Tong et al., "In-MRAM computing based on complementary-sensing time-based readout circuit using hybrid VGSOT-MTJ/GAA-CNTFET," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 72, no. 1, pp. 173–177, Jan. 2025.

[12] H. Zhang et al., "Spintronic processing unit within voltage-gated spin Hall effect MRAMs," *IEEE Trans. Nanotechnol.*, vol. 18, pp. 473–483, 2019.

[13] W. L. Peng et al., "Tunable damping-like and field-like spin-orbit-torque in Pt/Co/HfO_2 films via interfacial charge transfer," *Appl. Phys. Lett.*, vol. 115, no. 17, Oct. 2019, Art. no. 172403.

[14] G. G. B. Flores and K. D. Belashchenko, "Effect of interfacial intermixing on spin-orbit torque in Co/Pt bilayers," *Phys. Rev. B, Condens. Matter*, vol. 105, no. 5, Feb. 2022, Art. no. 054405.

[15] L. Jiang, E. Deng, H. Zhang, Z. Wang, W. Kang, and W. Zhao, "A spintronic in-memory computing network for efficient Hamming codec implementation," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 69, no. 4, pp. 2086–2090, Apr. 2022.

[16] B. Wu, H. Zhu, K. Chen, C. Yan, and W. Liu, "MLiM: High-performance magnetic logic in-memory scheme with unipolar switching SOT-MRAM," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 70, no. 6, pp. 2412–2424, Mar. 2023.

[17] X. Wang et al., "Large field-like spin-orbit torque and enhanced magnetization switching efficiency utilizing amorphous mo," *Nano Lett.*, vol. 24, no. 23, pp. 6931–6938, May 2024.

[18] Z. Wang, Z. Li, M. Wang, B. Wu, D. Zhu, and W. Zhao, "Field-free spin-orbit-torque switching of perpendicular magnetization aided by uniaxial shape anisotropy," *Nanotechnology*, vol. 30, no. 37, Jun. 2019, Art. no. 375202.

[19] C.-E. Fillion et al., "Gate-controlled skyrmion and domain wall chirality," *Nature Commun.*, vol. 13, no. 1, pp. 1723–2041, Sep. 2022.

[20] S. Li et al., "Ultralow power and shifting-discretized magnetic racetrack memory device driven by chirality switching and spin current," *ACS Appl. Mater. Interfaces*, vol. 15, no. 33, pp. 39946–39955, 2023.

[21] S. Li et al., "Interlayer Dzyaloshinskii–Moriya interaction in synthetic ferrimagnets," 2024, *arXiv:2403.12642*.

[22] D. Zhang et al., "Reliability-enhanced separated pre-charge sensing amplifier for hybrid CMOS/MTJ logic circuits," *IEEE Trans. Magn.*, vol. 53, no. 9, pp. 1–5, Sep. 2017.

[23] X. Li, X. Jin, W. Chen, N. Yin, and Z. Yu, "Parallel-prefix adder in spin-orbit torque magnetic RAM for high bit-width non-volatile computation," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 2, pp. 761–765, Feb. 2023.

[24] S. Yuasa et al., "Future prospects of MRAM technologies," in *IEDM Tech. Dig.*, Dec. 2013, pp. 3.1.1–3.1.4.

[25] L. Liu, C.-F. Pai, Y. L. H. W. Tseng, D. C. Ralph, and R. A. Buhrman, "Spin-torque switching with the giant spin Hall effect of tantalum," *Science*, vol. 336, pp. 555–558, Feb. 2012.

[26] A. Saha, S. Alla, and V. K. Joshi, "A novel time-domain in-memory computing unit using STT-MRAM," *Microelectronic Eng.*, vols. 284–285, Jan. 2024, Art. no. 112128.

[27] J. Wang et al., "Reconfigurable bit-serial operation using toggle SOT-MRAM for high-performance computing in memory architecture," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 69, no. 11, pp. 4535–4545, Nov. 2022.

[28] A. Fathollahi, A. Amirany, and M. H. Moaiyeri, "Ternary computing using a novel spintronic multi-operator logic-in-memory architecture," *Results Eng.*, vol. 25, Mar. 2025, Art. no. 104011.

[29] R. Zhou and H. Cai, "Time-domain computing for Boolean logic using STT-MRAM," *AIP Adv.*, vol. 13, no. 2, Feb. 2023, Art. no. 025102.

[30] Z. Tong et al., "A high throughput in-MRAM-computing scheme using hybrid p-SOT-MTJ/GAA-CNTFET," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 71, no. 2, pp. 606–619, Feb. 2024.

[31] Z. Wang et al., "Evaluation of ultrahigh-speed magnetic memories using field-free spin-orbit torque," *IEEE Trans. Magn.*, vol. 54, no. 11, pp. 1–5, Nov. 2018.

[32] Z. Wang, W. Zhao, E. Deng, J.-O. Klein, and C. Chappert, "Perpendicular-anisotropy magnetic tunnel junction switched by spin-Hall-assisted spin-transfer torque," *J. Phys. D, Appl. Phys.*, vol. 48, no. 6, Jan. 2015, Art. no. 065001.

[33] Z. Wang et al., "High-density NAND-like spin transfer torque memory with spin orbit torque erase operation," *IEEE Electron Device Lett.*, vol. 39, no. 3, pp. 343–346, Mar. 2018.

[34] M. Rezaei, A. Amirany, M. H. Moaiyeri, and K. Jafari, "A reliable non-volatile in-memory computing associative memory based on spintronic neurons and synapses," *Eng. Rep.*, vol. 6, no. 11, p. 12902, Apr. 2024.

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